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TRW Automotive

Slip Control Software Design

Document ID	:	ARM-DDD-Arm_Veh_State
Title	:	ARM-DDD-Arm_Veh_State
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Scope	:	Application: arm_veh_state{.c, _priv.h, _sub.h} Calibration:
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REFERENCES

Ref.	Document	Title	Ver.	Location

REVISION HISTORY

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1 ABSTRACT

This module calls the Active Roll Mitigation ([ARM](#)) functions for signal calculation.

2 ACRONYMS

ARM Active Roll Mitigation

ROI Roll Over Index

AyMod 'Modified' Lateral Acceleration, that tries to predict the Lateral Acceleration

3 OVERVIEW

This document describes the purpose and behavior of Arm_Veh_State.

3.1 DESIGN GOAL

ArmVehState calls all functions that are necessary to determine the state of the vehicle and to activate [ARM](#).

3.2 GENERAL REQUIREMENTS

1. [ARM](#) should prevent dynamic rollover.
2. [ARM](#) should prevent static rollover.

3.3 DETAILED REQUIREMENTS

Requirement ID	
Location	

3.4 SYSTEM MODE REQUIREMENTS

This module needs only to be processed, when [ARM](#) is allowed.

3.5 FLOW-CHARTS AND DIAGRAMS

3.5.1 Overview Arm Trigger Conditions

None.

4 DETAILED DESIGN

This chapter outlines the detailed design of Arm_Veh_State.

4.1 ALGORITHM DESCRIPTION

4.1.1 ArmVehState

In the function ArmVehState the functions ArmAyMod, ArmRollOverIndex and ArmState are scheduled. Detailed information on the latter functions can be found in the corresponding Detailed Design Documents.

- ARM-DDD-Arm_Ay_Mod
- ARM-DDD-Arm_Roll_Over_Index
- ARM-DDD-Arm_State

5 PROVIDED TASKS

5.1 ArmVehState

Description	ArmVehState
Rate	Standard Arm timebase 5 - 20ms
Dependency	
SYNTAX	void ArmVehState (void)

6 PROVIDED SERVICES

None.

7 CONFIGURATION OPTIONS

None.

8 CONFIGURATION PROCEDURES

None.

9 TEST CASES

Test category	
Requirements	
Test Case ID	

10 CALIBRATION INFORMATION

None.

11 APPENDIX